Guangdong Technion
Israel Institute of Technology
广东以色列理工学院

Differential and Integral Calculus 1 M - 104041

Exam A

January 18th, 2024

Your ID Number:

First name and Surname:

Guidelines

1. **Duration: 3 hours**. The use of personal dictionaries, electronic devices, reference materials, personal notes or any other extra material is not allowed.
2. Please, write your name and your ID number on the cover page.
3. The answers to the exam should be written in blue/black pen only. DO NOT use a pencil or correction fluid. DO NOT write near the staple area, otherwise it will be cut off.
4. Explain all your solutions and quote all the theorems you use. **No credits will be given for non-justified answers!** Write clear and complete answers for each problem on the same page.

1. (15p) Consider the following function $f(x)$, with a and b real numbers.

$$f(x) = \begin{cases} a\ln(x) + 2x & \text{if } x \geq 1, \\ \\ \dfrac{bx}{1+x^2}, & \text{if } x < 1. \end{cases}$$

(a) Find all the possible values for a and b such that $f(x)$ becomes continuous and differentiable on $\mathbb{R}$.

(b) Determine the local extreme values of the function on the interval $[-2, e]$.

(c) Determine the global extreme values of the function or explain why the function does not have global maximum or minimum values.

Answer

2. (20p) Consider the function

$$f(x) = \begin{cases} \dfrac{\cos(x)}{1+\sin(x)} & \text{if } \frac{-\pi}{2} < x \leq \pi \\[2ex] -\dfrac{1}{2}\left(\dfrac{\pi + x}{x}\right), & \text{if } \pi < x. \end{cases}$$

(a) Find all the vertical and horizontal asymptotes.

(b) Determine $f'(x)$, its domain and the intervals where the function increases and decreases.

(c) Find the inflection points and indicate the intervals where the function is concave upward and concave downward.

(d) Sketch the graph of the function.

Answer

3. (20p) Evaluate the following integrals

(a) $\displaystyle\int \cot(x)\left[\ln\left(\sin(x)\right)\right] dx$

(b) $\displaystyle\int_0^2 x^3 e^{x^2}\, dx.$

(c) $\displaystyle\int \frac{dx}{(6-x^2)^{3/2}}$

(d) $\displaystyle\int_1^\infty \frac{e^x}{1-e^{2x}} dx$

Answer

4. (10p) Find the length of the curve $y(x) = \displaystyle\int_1^x \sqrt{t^{2/3} - 1}\, dt$, for $1 \leq x \leq 8$.

<u>Answer</u>

5. (20p) (a) Determine the radius of convergence and the interval of convergence of the following power series.

$$\sum_{n=1}^{\infty} \frac{x^n}{n\sqrt{n}3^n}$$

(b) Consider the following function

$$f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!(2n+1)}.$$

Find its radius of convergence and show that $f(x) = \displaystyle\int_0^x \frac{\sin(t)}{t} dt$.

Answer

6. (15p) For each of the following statements answer TRUE or FALSE and justify your answer.

(a) Let $f(x)$ be a continuous function on the interval $[-3, 0]$ and differentiable on the interval $(-3, 0)$. If $f'(x) \leq 2$ for all $x \in (-3, 0)$ and $f(-3) = 5$, then $f(0)$ can take the value 15.

(b) $\displaystyle\lim_{x\to\infty} \left(1 + \frac{1}{x}\right)^{2x} = 1.$

(c) If g is a function such that $\displaystyle\int_0^{x^2} g(t)dt = 1 - e^{-2x^2}$, then $g(x) = 2e^{-2x^2}$.

<u>**Answer**</u>